

A 3D STRATEGY GAME FROM THE CAELUM
CHRONICLES UNIVERSE

3phor

KALEO EDITION

Where strategists gather.

THREE LAYERS · FOUR LIFTS · ONE NEXUS

RULEBOOK · v20 · CROSS-BOARD RULE

2 players · ages 8+ · 45–75 minutes

• Foreword

From SkyFlag to 3phor

This game began life in 2026 under the working title *SkyFlag*. The world, the pieces, the geometry, and the spirit of play have not changed; the brand has. The game you hold is now **3phor** — the inaugural game of the *Caelum Chronicles* universe, of which the Kaleo Edition is the first chapter.

Everything in the rulebook v19.1 (Cross-Board Rule, Edition 1.1) is preserved in v20. What's been added comes from a year of playing the game across solo, 2P hot-seat, and online formats — additions that *support* the core geometry rather than alter it.

What's new in v20

ADDITION	EFFECT ON THE CORE GAME
Time controls (5 / 10 / 30 min)	Optional. Off by default; identical core rules.
Resignation & draw offers	Either player may resign or offer a draw at any time.
AI difficulty levels	Easy / Medium / Hard / Expert. Solo play only.
Online multiplayer	Rated games, friends, tournaments. Same rules.
Daily Puzzle	One shared tactical position per day. Single-side, race-the-clock.

• Kaleo

A world above a world that failed

Long ago, when the planet below gave way to salt, dust, and ruined basins, the Aetheri lifted what remained of civilization into a three-layer arcology. The lowest layer — the Terran — holds the reservoirs, farms, workshops, and crowded human districts. Above it, the Meridian carries charged air, transit architecture, and the old routing systems. At the summit rests the Empyrean — the quiet upper shell where the Aetheri left their highest instruments.

At the centre of the Empyrean, and only there, waits the Caelum Nexus. The lower layers have no such central shrine. What they have, instead, are corners.

Each clan placed three dormant claim-seals in the corners of its own territory on each layer — proofs of presence left for the Aetheri to witness. Defend your seals; the opponent's seals are your targets. Only by crossing into the opposing clan's territory and neutralizing all three of its seals may a Captain earn the right to approach the Nexus. The Aetheri did not trust survival to strength alone; they built Kaleo as a system of proofs, and the proof demanded is reach.

“The Aetheri did not build three layers. They built three answers to the same question.”

The Two Clans

For seven generations, Kaleo was one people. The schism came during a crisis whose name the archives still record — a disease that tore through the lower districts while the council debated what to do. One faction argued for immediate escalation: open the Aetheri archives, petition the Nexus. The other argued for containment: preserve what remained, refuse panic, work with the tools already at hand.

The containment faction won the argument by one vote. Their victory stabilized Kaleo at the cost of two hundred thousand Terran residents who might have been saved. The escalation faction did not forget. Over four generations they consolidated into the **Grey Ravens**. The faction that had won became the **White Stags**. Each clan drew its name and totem from the temperament that had defined its position in that ancient debate.

Six generations later, both clans still claim duty. Both claim restraint. Both claim they are acting for Kaleo. Both are correct. Neither is complete. The contest that decides Kaleo's next age is the raid on

the opposing clan's three seals, and the answer of the Nexus.

GREY RAVENS

"We have been here longer than the crisis. We will be here longer than its answer."

WHITE STAGS

"We were given a path. If we do not walk it, it becomes a monument to our refusal."

• Contents

This rulebook proceeds in the order you will need it. Pages marked **NEW** are additions in v20.

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HOW TO USE THIS RULEBOOK

If you are sitting down to your first game, read sections 1–4 before setup. Section 10 is a single-page reference designed to live beside the board during play.

1 Overview

3phor is a two-player tactical race played across a three-layer arcology. Each player commands four pieces — a Captain, a Soldier, a Rover, and a Pilot — and pursues the opposing clan's three claim-flags, one on each layer of the board. Flags may be captured in any order. Once all three of the opponent's flags are claimed, your Captain must land on the **Caelum Nexus** at Space(3,3) to win.

Pieces move and capture by geometry alone. There are no special abilities, no cards, no dice. Everything the game offers comes from the interaction of four movement profiles with the architecture of three stacked boards and the four vertical transit points — the **Lifts** — that connect them.

THE OBJECTIVE

Each player must capture the opponent's three flags, one on each layer of the arcology. Capture all three — in any order — then land your Captain on the Nexus at Space(3,3) to win.

Board coordinates

All three layers share the same coordinate system. A cell is identified as *Layer(row, col)*. Row 0 is the top edge; row 5 is the bottom. Col 0 is the left edge; col 5 is the right.

LAYER NAME	IN-WORLD NAME	NOTABLE FEATURE
Ground	Terran	Starting layer — pieces deploy here.
Sky	Meridian	Middle transit layer. Most contested phase.
Space	Empyrean	Contains the Nexus at Space(3,3).

2 Components & Setup

Components

ITEM	QUANTITY & NOTES
Three-layer board	Ground, Sky, Space — each a 6×6 grid sharing coordinates.
Player pieces	4 per player: Captain, Soldier, Rover, Pilot. P1 = slate; P2 = ivory.
Flag markers	3 per player: Ground Flag (▢), Sky Flag (▢), Space Flag (★).
Lift tokens	12 total — four per layer, marking the Lift cells.
Nexus marker	1 — placed at Space(3,3).

The Board

The board comprises three stacked 6×6 layers. Four Lifts at (1,1) · (1,4) · (4,1) · (4,4) connect every layer. Flag positions are fixed and symmetric for the two players.

GROUND — THE TERRAN LAYER

	c0	c1	c2	c3	c4	c5
r0	▢ P1			● P1		
r1		↑			↑	
r2						
r3						
r4		↑			↑	
r5			● P2			▢ P2

P1 Ground Flag at G(0,0); P2 at G(5,5). Captains start: P1 G(0,3), P2 G(5,2). Lifts ↑ at (1,1), (1,4), (4,1), (4,4).

Sky — the Meridian layer

	c0	c1	c2	c3	c4	c5
r0						▮ P1
r1		↑			↑	
r2						
r3						
r4		↑			↑	
r5	▮ P2					

P1 Sky Flag at Sky(0,5); P2 at Sky(5,0) — diagonally opposite.

Space — the Empyrean layer

	c0	c1	c2	c3	c4	c5
r0	★ P1					
r1		↑			↑	
r2						
r3				◎ NX		
r4		↑			↑	
r5						★ P2

P1 Space Flag at Space(0,0); P2 at Space(5,5). Each echoes the Ground Flag corner (the Aetheri Echo Point).

The Nexus ◎ sits at Space(3,3).

LIFTS — ALL LAYERS

The four Lift positions are the same on all three layers: $(1,1) \cdot (1,4) \cdot (4,1) \cdot (4,4)$. No two Lifts are within king-move of each other. A Captain cannot threaten two Lifts from a single position — Lift defence is by positioning, not cluster control.

Setup

	P1 · SLATE	P2 · IVORY	NOTES
Captain start	Ground (0,3)	Ground (5,2)	In hand; deploy costs 1 activation.
Soldier	In hand	In hand	Deploy during play — 1 activation.
Rover	In hand	In hand	Deploy during play — 1 activation.
Pilot	In hand	In hand	Deploy during play — 1 activation.
Ground Flag 🚩	G(0,0)	G(5,5)	Place at each player's corner.
Sky Flag 🚩	Sky(0,5)	Sky(5,0)	Opposite corner on the Sky layer.
Space Flag ★	Space(0,0)	Space(5,5)	Echoes the Ground Flag corner.
Nexus ◎	Space(3,3)	Space(3,3)	Shared finish cell.
Activations	2 per turn	2 per turn	Both players start with the same budget.

SETUP STEPS

1. Place 12 Lift tokens — 4 per layer at (1,1), (1,4), (4,1), (4,4).
2. Place P1's flags at G(0,0), Sky(0,5), Space(0,0). Place P2's flags at G(5,5), Sky(5,0), Space(5,5).
3. Place the Nexus marker ◎ at Space(3,3).
4. All pieces (Captain, Soldier, Rover, Pilot) start in hand. Determine the first player by any mutual method.
5. Each turn, both players have 2 activations.

3 The Four Pieces

All tactical power comes from movement, capture geometry, and positioning. Pieces use the same movement on every layer. The summary table below is followed by a detailed card for each piece.

PIECE	MOVEMENT	CAPTURE	ROLE
Captain ♔	King-move: 1 sq in any of 8 directions.	Lands on opponent's cell.	Win-capable. Captures flags & lands on Space(3,3) to finish.
Soldier ♟	Forward only. First move ≤ 2 sq; subsequent 1 sq.	1 sq diagonally forward.	Deploys from hand. Promotes to Captain at the far row on Ground.
Rover ♖	Orthogonal ≤ 2 sq.	1 step only — cannot capture at 2 squares.	Lift-corridor guard. Blocks approaches.
Pilot ♙	Diagonal ≤ 2 sq. May jump over an intervening piece.	Lands on opponent beyond the jump — jumped piece captured.	Flanks Lift approaches. Reaches threats over defenders.

CAPTAIN • DANTEC

Slate — Renn Dantec · Ivory — Sera Dantec

The only piece — along with a promoted Soldier — that can capture flags and answer the Nexus. Every other piece exists to protect her approach.



MOVEMENT	1 sq in any of 8 directions (king-move). All layers.
CAPTURE	By landing on the opponent's cell.
ROLE	Win-capable. Captures flags. Finishes the game at Space(3,3).

SOLDIER · DURREN

Slate — Toren Durren · Ivory — Kael Durren

Deploys from hand. Forward-only. A Soldier that reaches the back rank on Ground changes the game: a second win-capable piece appears on the board.



MOVEMENT

Forward only (P1 down rows; P2 up rows). First move ≤ 2 sq; subsequent 1 sq.

CAPTURE

1 sq diagonally forward only.

ROLE

Controls approach corridors. Promotes to Captain at the far row.

SOLDIER PROMOTION

When a Soldier reaches the far row on the Ground layer (row 5 for P1; row 0 for P2), it promotes immediately to a Captain. A promoted Captain has full Captain powers: it may capture flags and land on Space(3,3) to finish the game.

Flag-capture progress is shared across both Captains — you do not need to capture all three flags with the same Captain. Players may therefore field up to two Captains simultaneously.

ROVER · THANDIWE

Slate — Mox Thandiwe · Ivory — Iva Thandiwe

Rovers lock down Lift corridors and force the opponent to reroute. A Rover does not need to strike — a Rover only needs to stand in the correct place.



MOVEMENT

Orthogonal ≤ 2 squares (horizontally or vertically). All layers.

CAPTURE

1 step only — cannot capture at 2 squares.

ROLE

Limited rook. Corridor control. Lift defence.

PILOT · VOSS

Slate — Zara Voss · Ivory — Arin Voss

The Pilot converts a screen into a liability. The one piece that reaches a threat over defending pieces — arriving one square sooner than the defender expected.



MOVEMENT

Diagonal ≤ 2 squares. May jump over an intervening piece.

CAPTURE

Lands on opponent beyond the jump — jumped piece is captured.

ROLE

Limited bishop with jump-capture. Flanks Lift approaches.

PILOT JUMP DETAILS

A Pilot at A may jump to C if there is exactly one piece (friend or enemy) at B, where $A \rightarrow B \rightarrow C$ is a straight diagonal of 2 squares. If C holds an enemy piece, both that piece *and* the jumped piece at B are captured.

If C is empty, the move is legal — the Pilot still passes over the jumped piece but only the jumped piece is captured.

A Pilot cannot jump over more than one piece, and cannot end its move on a square occupied by a friendly piece.

4 Turn Sequence

Each turn proceeds through the following steps in order. You must take every action available within your activation budget — you cannot pass if a legal action exists.

STEP	WHAT YOU DO
1 • Deploy (optional)	Place 1 piece from hand onto your starting cell on the Ground layer. Costs 1 activation. A newly deployed piece cannot move on the same activation.
2 • Activate	Take up to 2 actions with deployed pieces. Each action is one of: move a piece, capture, step up or down a Lift. You must act if any legal action is available.
3 • Flag check	If one of your Captains (original or promoted) just landed on a cell containing an opponent's flag, capture it immediately. Remove the flag marker.
4 • Nexus check	If all three of your opponent's flags have been captured and one of your Captains just landed on Space(3,3), you win immediately.
5 • Pass initiative	The other player now takes their turn.

ACTIVATION SUMMARY

Standard: 2 activations per turn for both players.

Lift use: move to a Lift cell (1 activation on one turn), then step up or down one layer (1 activation on the next turn). The Lift step itself cannot capture.

Flag capture is automatic at step 3 — it does not consume an activation, and it grants no bonus activation.

5 The Lifts

Lifts connect the three layers. They always occupy the same (row, col) on each layer: (1,1) · (1,4) · (4,1) · (4,4). These four positions form a square ring inside the board, each exactly one king-move from a flag corner on the Ground layer.

USING A LIFT — TWO TURNS

Using a Lift takes two separate activations on two separate turns:

- **Turn A:** move your piece onto the Lift cell on its current layer (1 activation).
- **Turn B:** step your piece up or down one layer via that Lift (1 activation).

A piece cannot arrive at a Lift and step through it in the same activation. *Exception:* if a piece ends a turn already on a Lift cell, it may step up or down on its very next activation.

LIFT OCCUPATION RULE

A piece on a Lift cell on one layer blocks the Lift step from the adjacent layer. The blocking piece does *not* need to be on the same Lift cell as the ascending piece — it needs to be at the same (row, col) on the layer above or below.

The blocked Lift step cannot capture the blocking piece. To clear a blocked Lift, remove the blocking piece through normal play on its own layer.

Strategy: parking a defender on the Space Lift above your own flag (P1: Space(1,1); P2: Space(4,4)) simultaneously guards the approach AND blocks that Lift from Sky, denying the opponent an ascent.

Lift independence: no two Lifts are within king-move of each other. A piece on one Lift cell cannot threaten any other Lift. There is no cluster fortress — each Lift must be defended individually through positioning.

6 The Flags & the Nexus

Each player captures the opponent's three flags, placed in the opponent's corners. Landing on your own flag cell does nothing — your flag stays in place until the opponent captures it. Flags may be captured in any order — Ground, Sky, and Space are fully independent objectives. Once all three of the opponent's flags are captured, your Captain must reach the Nexus at Space(3,3) to win.

	P1 · SLATE (TARGETS)	P2 · IVORY (TARGETS)
Ground Flag 🏴	G(5,5) — bottom-right	G(0,0) — top-left
Sky Flag 🏴	Sky(5,0) — bottom-left	Sky(0,5) — top-right
Space Flag ★	Space(5,5) — bottom-right	Space(0,0) — top-left
Nearest Lift	(4,4) for G & Space · (4,1) for Sky	(1,1) for G & Space · (1,4) for Sky
Nexus ☉	Space(3,3) — shared finish cell	Space(3,3) — shared finish cell

FLAG CAPTURE

A Captain (original or promoted) lands on an opponent's flag cell → the flag is captured instantly. Remove the marker. A friendly or enemy piece on the flag cell does *not* block capture — a Captain may always land on and capture an opponent's flag regardless of who else occupies it.

Flag capture does not grant a bonus activation. The turn continues with its normal budget.

After all three opponent flags are captured, your Captain must reach Space(3,3) to win.

The Aetheri Echo Point

Each player's Ground and Space flags share the same (row, col) — the *Aetheri Echo Point*. Continuity of position across layers was the proof the Aetheri required. The Sky flag sits at the diagonally opposite corner, enforcing a genuine horizontal traverse on the central layer.

7 Win Conditions

CONDITION	HOW IT TRIGGERS
Win by Nexus (primary)	A Captain on your side captures all three of your opponent's flags — Ground, Sky, and Space, in any order — and then lands on the Nexus cell at Space(3,3). Either the original Captain or a promoted Captain may trigger this win. Flags may be captured across both Captains collectively.
Win by Elimination	Your opponent has no Captain on the board and none in hand, and no Soldier remaining that could promote to a Captain. All Captain-capable pieces have been permanently captured.
Win by Resignation NEW	Your opponent resigns the game from the in-game toolbar. The win is recorded as a resignation in the stats.
Win by Time-out NEW	If a time control is in effect and your opponent's clock reaches zero, you win immediately. Time controls are optional and default to off.
Draw by Agreement NEW	Either player may offer a draw at any time; if accepted, the game ends with no winner.
Turn-limit Draw	180 turns without a win. Tiebreak, in order: (1) flags captured; (2) pieces remaining; (3) opponent pieces captured; (4) highest layer occupied; (5) Captain Chebyshev distance to Space(3,3).

Competitive Note

3phor contains a minor structural asymmetry in its spatial geometry. Specifically, Space(3,3) is not equidistant from the Player 1 (P1) and Player 2 (P2) starting positions when evaluated through optimal Captain movement paths. Under efficient play, P1 can reach this key space in 18 actions, whereas P2 requires 19. This one-action differential introduces a slight first-player advantage, analogous to the initiative observed for White in Chess. To mitigate this imbalance in competitive play, matches are structured as a best-of-three series, with players alternating starting positions between games. This format ensures that each player has equal exposure to both positional advantages over the course of the match.

8 Game Phases

The three-flag system with a Nexus finish creates four natural phases of play. Flags may be pursued in any order, but the geometry rewards a sequential path.

PHASE	LAYER	WHAT HAPPENS
1 • Ground Race	Ground	Both Captains on Ground racing across the board to capture the opposing Ground Flag. Soldiers deploy during play. Rovers and Pilots may ascend to Sky or Space immediately. Early Lift skirmishes determine approach control. ~3–4 rounds.
2 • Sky Crossing	Sky	Sky transit is the hardest leg: the target Sky Flag is diagonally opposite from the Lift nearest your Ground approach (four king-moves). Both Captains must cross the full board width in Sky. Most contested phase. ~4–5 rounds.
3 • Space Flag	Space	After Sky, the Captain ascends to Space. With echoed corners, the target Space Flag sits at the same position as the target Ground Flag — immediate recognition, but the opponent may already be contesting the layer. ~2–3 rounds under competition.
4 • Nexus Finish	Space	With all three flags captured, the Captain must converge on Space(3,3). The Nexus is central — 2 or 3 king-moves from any Space Flag — and either Captain (original or promoted) can finish. 2–3 activations minimum.

9 Play Modes NEW IN V20

3phor is the same game in every mode. These mode descriptions document how it's actually played on the digital editions (3phor.com on web and iOS); the rules above govern all of them.

Solo vs AI

Single-player against a computer opponent. The AI uses an iterative-deepening minimax search with alpha-beta pruning, transposition table, killer-move + history heuristic move ordering, a piece-square table evaluator, and a small opening book. Four difficulty levels:

DIFFICULTY	SEARCH DEPTH	BEST FOR
Easy	2 ply	First-time players learning the geometry.
Medium	3 ply	Once you've won a few Easy games.
Hard	4 ply	Default for confident players.
Expert · <i>Plus</i>	6 ply	Sterner opponent. Requires the Plus subscription.

Difficulty is set in *Settings* → *AI difficulty*. The choice persists across sessions.

Two-Player Hot-Seat

Two humans sharing one device. Pass after each turn. No clock by default; enable a clock from *Settings* → *Clock* for a timed game. No account required.

Online Multiplayer

Two humans on separate devices, real-time over the network. Both players sign in (guest, email, Apple, or Google) and either create a room or join one by code. Sessions sync via Supabase realtime; push notifications fire on your turn if the app isn't focused.

- **Ratings.** Every completed online game updates both players' ELO ratings. K-factor 24.
- **Friends.** Save opponents' nicknames; see who's online and challenge directly.
- **Tournaments.** Round-robin and bracketed events. Free events open to all; entry-fee events require the Tournament Entry entitlement.

Daily Puzzle

One shared tactical position per day. The same starting position is served to every player worldwide; solve as efficiently as you can. A briefing screen presents the objective and a short rules recap before the position is revealed (the chess.com / Lichess "I'm ready" pattern).

Time Controls

Optional. Default off. When a clock is in effect, each player has the chosen budget for the entire game — there is no per-move increment. If your clock reaches zero, you lose by Time-out.

OPTION	PER SIDE	BEST FOR
Off	—	Untimed casual play.
5 minutes	5 min	Blitz — fast, decisive.
10 minutes	10 min	Rapid — most online matches.
30 minutes	30 min	Classical — thoughtful play.

Resignation & Draw Offers

Either player may resign at any time from the in-game toolbar. Either may also offer a draw; the opponent accepts or declines. A resignation is recorded as a loss for the resigner; an accepted draw counts as a draw for both. In solo play, the AI's "willingness" to accept a draw depends on the position evaluation — if the AI believes it is winning, it will decline.

10 Quick Reference

Keep this page beside the board during play.

RULE	SUMMARY
Win	Captain captures all 3 opponent flags (Ground, Sky, Space) in any order, then lands on Space(3,3).
P1 targets	G(5,5) · Sky(5,0) · Space(5,5)
P2 targets	G(0,0) · Sky(0,5) · Space(0,0)
Nexus ☉	Space(3,3) — shared finish cell.
Lifts	(1,1) · (1,4) · (4,1) · (4,4). Same on all layers. Two-turn transit. Not within king-move of each other.
Lift occupation	Piece above or below blocks the Lift step from the adjacent layer. The step cannot capture.
Activations	2 per turn. Flag capture does not grant a bonus.
Captain ♚	King-move (1 sq, 8 dirs, all layers). Capture by landing. Flag-capable. Wins on Space(3,3).
Soldier ♙	Forward-only. First move ≤ 2 sq. Capture: diagonal-forward. Promotes to Captain at the far row on Ground.
Rover ♖	Orthogonal ≤ 2 sq. Capture at 1 step only.
Pilot ♗	Diagonal ≤ 2 sq. May jump an intervening piece to capture beyond.
Promotion	Soldier reaching far row on Ground becomes a Captain. Can capture flags and trigger the Nexus win.
Any layer	Any piece may ascend or descend via Lift at any time.
Turn limit	180 turns. Tiebreak: flags > pieces > captures > layer > distance to Space(3,3).
Clock NEW	Optional 5 / 10 / 30 minutes per side. Zero clock = Time-out loss.

Resign / Draw

NEW

Either player may resign or offer a draw at any time.

3 phor · Kaleo Edition · Rulebook v20 · Cross-Board Rule

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